

HW0 LA Review in R

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```
library('dplyr')
data <- read.table("BirthWeights.txt", sep = " ", header = TRUE, as.is = FALSE)
```

1.

```
# Design matrix
X <- model.matrix(~ Mage + Gage + Race + Gen, data = data)
y <- matrix(data[, 1], ncol = 1)

# beta = (X'X)^(-1)*X'y
beta <- solve(t(X) %*% X) %*% t(X) %*% y

# S2
n <- nrow(X)
p <- ncol(X)
resids <- y - (X %*% beta)
S2 <- sum(resids^2) / (n-p)

lm <- lm(BirthWeight ~ ., data = data)

beta
```

```
              [,1]
(Intercept) -4120.542409
Mage         -3.793751
Gage         182.742497
Racehisp     198.747954
Raceother    241.582827
Racewhite    204.888197
GenMale     -169.348562
```

```
lm$coefficients
```

(Intercept)	Mage	Gage	Racehisp	Raceother	Racewhite
-4120.542409	-3.793751	182.742497	198.747954	241.582827	204.888197
GenMale					
-169.348562					

```
S2
```

```
[1] 79277.09
```

```
summary(lm)$sigma^2
```

```
[1] 79277.09
```

2.

```
# Manual vs. LM
table(round(X %*% beta, 3) == round(lm$fitted.values, 3))
```

```
TRUE
832
```

3.

```
# Manual vs. LM
table(round(y - X%*%beta, 3) == round(lm$residuals, 3))
```

```
TRUE
832
```